PAST PAPER SPRING 2024

SHORT QUESTIONS.

1- Define coordination number ----

Coordination number refers to the number of atoms, ions or molecules directly bounded to a central atom or ion in a crystal structure

⇒ It indicates the number of nearest neighbours to the central atom or ion

Coordination no. of	SC	6
Coordination no. of	BCC	8
Coordination no. of	FCC	12

2- ~~D~~ What are symmetry operations? Define each briefly?

Symmetry operations are transformations that can be performed on an object or a molecule, leaving it in an indistinguishable or identical configuration as before the operation

⇒ These operations are fundamental in understanding molecular symmetry, crystal structures & group theory.

Types of Symmetry operations

1- Identity (E) operation
2- Rotation (C_n) operations
3- Reflection (σ) operations

Identity (E)

- The simplest operation where nothing changes.
- Every object or molecule possesses this symmetry

Rotation (C_n)

- Involves rotating an object around an axis by a specific angle such that it appears identical.
- The axis is called axis of rotation

Example:- A benzene molecule has 6 axis of rotation (C_6)

Reflection (σ)

- A reflection through a mirror plane.

3- State laue diffraction conditions & explain all terms involved in it

Laue diffraction conditions describe the diffraction of x-rays by a crystal lattice. The laue equations are

(1) $a \cdot (k - k_0) = m$

(2) $b \cdot (k - k_0) = n$

(3) $c \cdot (k - k_0) = p$

Terms Involved

- a, b, c are the primitive lattice vectors of the crystal lattice

⇒ They represent the fundamental translation vectors that define the crystal structure.

- k & k_0 are the wave vectors of diffracted & incident waves.

⇒ Wave vectors are the vectors that describe the direction & magnitude of a wave.

- m, n & p are the integers th represent the order of diffraction.

$\Rightarrow$ They describe the number of wavelengths that fit within the crystal lattice.

4- The interplanar spacing d_{hkl} in a real space is equal to $\frac{2\pi}{|G|}$

Here $|G|$ represent the reciprocal lattice

$$|G| = (h\vec{a}^* + k\vec{b}^* + l\vec{c}^*)$$

$$|G| \cdot n = (h\vec{a}^* + k\vec{b}^* + l\vec{c}^*)\hat{n}\cos\theta$$

For $\theta = 0°$ (maximum)

magnitude of unit vector $\hat{n} = 1$

$$|G| \cdot n = (h\vec{a}^* + k\vec{b}^* + l\vec{c}^*)$$

$$n = \frac{(h\vec{a}^* + k\vec{b}^* + l\vec{c}^*)}{|G|} \quad \text{—— (1)}$$

Now, interplanar spacing is d_{hkl}

For x-axis

$$d_{hkl} = \frac{a}{h}$$

For y-axis

$$d_{hkl} = \frac{b}{k}$$

For z-axis

$$d_{hkl} = \frac{c}{l}$$

For x-axis

$$d_{hkl} = \frac{a}{h}$$

$$d_{hkl} = \frac{a}{h} \cdot n \quad \text{(put value of } n)$$

$$d_{hkl} = \frac{a}{h}\left[\frac{h\vec{a}^* + k\vec{b}^* + l\vec{c}^*}{|G|}\right]$$

$$= \frac{\cancel{\frac{h}{h}}(\vec{a}\cdot\vec{a}^*) + \frac{k}{h}(\vec{a}\cdot\vec{b}^*) + \frac{l}{h}(\vec{a}\cdot\vec{c}^*)}{|G|}$$

$$= \frac{2\pi + 0 + 0}{|G|} \qquad [\vec{a}\cdot\vec{a}^* = 2\pi]$$

$$\boxed{d_{hkl} = \frac{2\pi}{|G|}}$$

proved

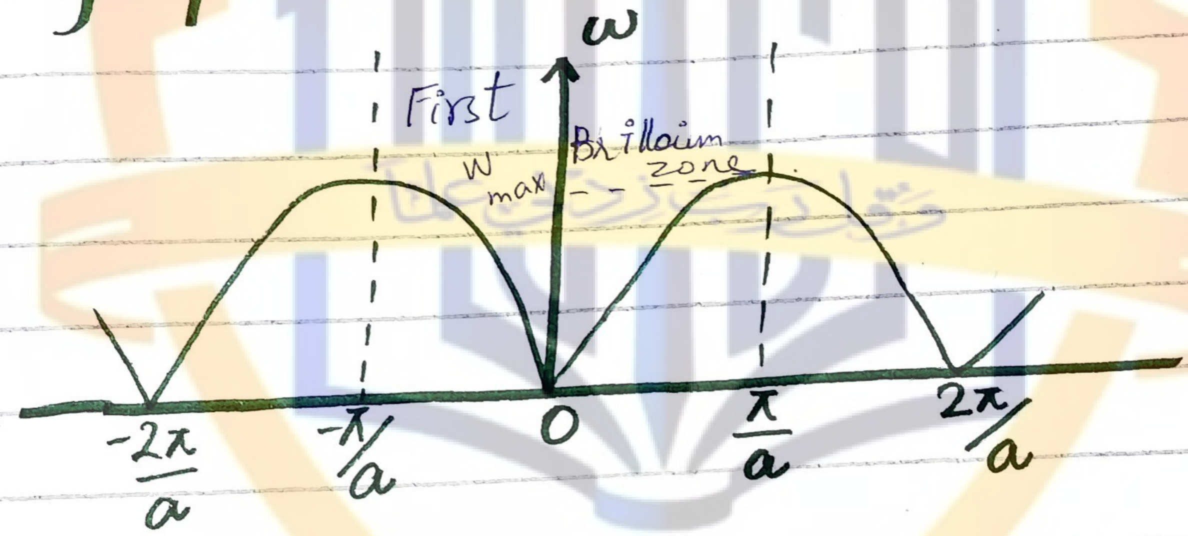
Imp S/Q (e)

If we consider only its magnitude then

$$\boxed{\omega = 2\sqrt{\frac{\beta}{m}}\sin\frac{ak}{2}}$$

The relation b/w 'ω' & 'k' is called Dispersion relation.

Graph between 'ω' & 'k'



for $k \to 0$ $\omega = 0$ $\lim_{k\to 0} \sin\frac{ka}{2} \sim \frac{ka}{2}$

As $\omega_z = \sqrt{\frac{4\beta}{m}}\,\frac{a}{2}\,k$

↓

constant

$\omega \propto k$

$\cancel{\sin}\frac{ka}{2} = \cancel{\sin}\frac{\pi}{2}$

$\frac{ka}{\cancel{2}} = \frac{\pi}{\cancel{2}}$

$ka = \pi$

$k = \pi/a$

Imp S/Q

For maximum frequency

$$\left|\sin\left(\frac{ka}{2}\right)\right|_{max} = 1$$

$$\omega_{max} = 2\sqrt{\frac{\beta}{m}}$$

$$\cancel{2}\pi V_{max} = \cancel{2}\sqrt{\frac{\beta}{m}} \qquad w_{max} = 2\pi V_{max}$$

$$V_{max} = \frac{1}{\pi}\sqrt{\beta/m}$$

For longer wavelength

In longer wavelength

$$K \to 0 \quad \& \quad \lambda = \frac{2\pi}{k}$$

As we have

$$W = 2\sqrt{\beta/m}\,\sin\left(\frac{ka}{2}\right)$$

It means angle is very small.

$$\sin\left(\frac{ka}{2}\right) \approx \frac{ka}{2}$$

$$W = \cancel{2}\sqrt{\frac{\beta}{m}}\,\frac{ka}{\cancel{2}}$$

$$\boxed{W = a\sqrt{\frac{\beta}{m}}\,k} \quad — Ⓐ$$

For Phase Velocity

$$\frac{\omega}{k} = a\sqrt{\frac{\beta}{m}}$$

$$\boxed{V_p = a\sqrt{\frac{\beta}{m}}}$$

Hence phase velocity becomes constant.

at longer wavelength or low frequency

For group velocity

from Ⓐ

$$d\omega = a\sqrt{\frac{\beta}{m}}\,dk$$

$$\frac{d\omega}{dk} = a\sqrt{\frac{\beta}{m}}$$

$$\boxed{V_g = a\sqrt{\beta/m}}$$

which is also constant.

$$\boxed{V_p = V_g}$$

Long Questions 2 (a)

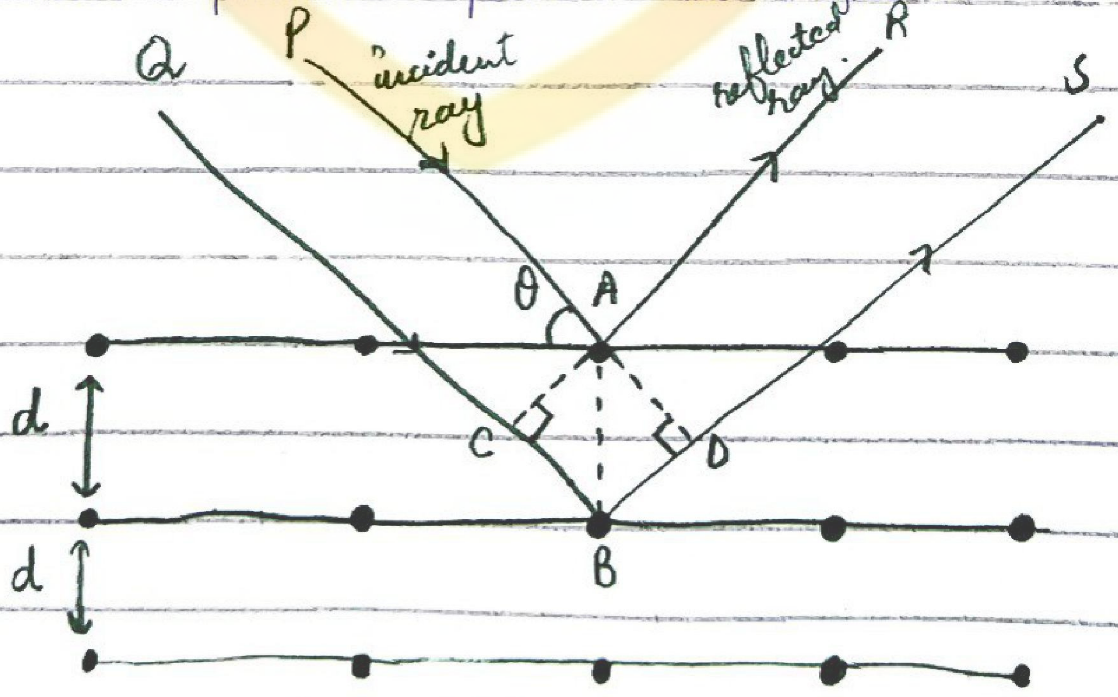
Bragg's law

$$n\lambda = 2d\sin\theta$$

In 1912, W.L bragg gave a simple formula for geometrical condition which must be satisfied if x-rays are to be diffracted by parallel set of planes. This simple formula is known as bragg's law.

$$n\lambda = 2d\sin\theta$$

Consider a set of parallel planes. Let the planes be at distance d from each other. Let a parallel beam of x-rays of wavelength λ be incident on these parallel planes at angle θ.



Draw $AC \perp QB$ & $AD \perp BS$ to find the path difference

The reflected rays AR & BS will interface constructively if the path difference CB + BD is equal to integral multiple of λ

$$CB + BD = n\lambda \quad \text{——} \quad (1)$$

In ΔACB,

$$\sin\theta = \frac{CB}{AB} \qquad [\because AB = d]$$

$$\sin\theta = \frac{CB}{d}$$

$$CB = d\sin\theta \quad \text{——} \quad (2)$$

In ΔADB,

$$\sin\theta = \frac{BD}{AB}$$

$$\sin\theta = \frac{BD}{d} \qquad [\because AB = d]$$

$$d\sin\theta = BD \quad \text{——} \quad (3)$$

Adding equ (2) & (3)

$$CB + BD = d\sin\theta + d\sin\theta$$

$$CB + BD = 2d\sin\theta \quad \text{——} \quad (4)$$

Importance of bragg's law

1- Band stucture Analysis
2- Crystal structure determination
3- Defects & Imperfections
4- Material classification
5- Development of advanced materials.

Vector form of bragg's law

$$\vec{n} \cdot \vec{d} = n\lambda$$

- $\vec{n}$ is the diffraction vector (the vector that points from origin to the point of diffraction in reciprocal space)
- $\vec{d}$ is the vector normal to the crystal planes (perpendicular to the planes)
- n is an integer representing the order of diffraction
- λ is the wavelength of the incident wave

Comparing equation 1 & 4

$$\boxed{n\lambda = 2d\sin\theta}$$

which is required bragg's law equation

Physical Significance

Bragg's law helps us understand & analyze the arrangement of atoms in a crystal by studying the patterns formed when x-ray bounce off the crystal's atomic layers.

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Question 2(b)

Quasi Crystals

Quasi crystals are a unique type of solid material in which the atoms are arranged in an ordered structure, but unlike regular crystals, this arrangement does not repeat periodically.

⇒ Instead, Quasicrystals exhibit quasiperiodic order, meaning their or patterns are ordered but never exactly repeat.

Physical properties:

1- Low thermal conductivity

2- High melting point

3- High Hardness

4- Unusual electrical properties

Applications

Used in
- Non-stick coatings
- LED Lights
- Reinforcement material due to their strength

- Thermal barrier materials
- Wear-Resistant coatings
- Decorative applications.

Examples

- Penrose tiling (mathematical method)
- Icosahedrite (Al_6Fe_2Cr)

Einstein Model.

Question 3(a)

Einstein ~~Model~~ was a first person to resolve the problem of specific heat of solid by the application of Planck's Quantum Theory.

He assumed that a crystal solid is made up of N atoms (per mole) could be regarded as an array of atomic oscillators vibrating in three diff independent directions (as assumed for classical calculation) with a constant frequency ν_0. The energies of these oscillators are not continuous but discrete and are given by

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega$$

If we neglect zero point energy then we may write

$$E_n = n\hbar\omega$$

Where the quantum number $n = 0, 1, 2 \ldots$

$\hbar = \frac{h}{2\pi}$ and $\omega = 2\pi\nu$

The number of oscillators N_n

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of each energy state is determined from the Boltzmann function as

$$N_n = N_0 \exp\left(\frac{-E_n}{K_B T}\right)$$

The average energy $= \frac{E}{N} = \frac{\text{Total energy}}{\text{No. of oscillators}}$

$$\bar{E} = \frac{\sum_n N_n E_n}{\sum_n N_n}$$

Where N_n is the Number of oscillator in n^{th} state.

$$\bar{E} = \frac{\sum_n N_0 \exp\left(\frac{-E_n}{K_B T}\right) n\hbar\omega}{\sum_n N_0 \exp\left(\frac{E_n}{K_B T}\right)}$$

$$\bar{E} = \frac{\cancel{N_0}\,\hbar\omega \sum_n n \exp\left(-\frac{n\hbar\omega}{K_B T}\right)}{\cancel{N_0} \sum_n \exp\left(\frac{-n\hbar\omega}{K_B T}\right)}$$

$$\bar{E} = \frac{\hbar\omega \sum_n n \exp\left(-\frac{n\hbar\omega}{K_B T}\right)}{\sum_n \exp\left(\frac{-n\hbar\omega}{K_B T}\right)}$$

Let $\exp\left(\frac{-n\hbar\omega}{K_B T}\right) = x^n$

$$\bar{E} = \frac{\hbar\omega \sum_n n x^n}{\sum_n x^n}$$

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$\because \sum_s x^s = \frac{1}{1-x}$, $\sum_s s x^s = x\frac{d}{dx}\sum_s x^s$

So,

$$\bar{E} = \frac{\hbar\omega \, x\frac{d}{dx}\sum_n x^n}{\sum_n x^n} , = \frac{\hbar\omega \, x\frac{d}{dx}\left(\frac{1}{1-x}\right)}{1/1-x}$$

$$\bar{E} = (1-x)\hbar\omega x(-1)(1-x)^{-2}(-1)$$

$$\bar{E} = \frac{(1-x)\hbar\omega x}{(1-x)^2} , \quad \bar{E} = \frac{\hbar\omega x}{1-x}$$

$$\bar{E} = \frac{\hbar\omega \exp\left(-\frac{\hbar\omega}{k_BT}\right)}{1-\exp\left(-\frac{\hbar\omega}{k_BT}\right)}$$

$$\bar{E} = \frac{\hbar\omega}{e^{\frac{-\hbar\omega}{k_BT}}-1}$$

The total energy of one mole of a solid having 'N' atoms is therefore

$$E = 3N\bar{E}$$

$$E = \frac{3N\hbar\omega}{\exp\left(\frac{\hbar\omega}{k_BT}\right)-1}$$

and $C_V = \left(\frac{\partial E}{\partial T}\right)_V$

$$\frac{\partial E_{mol}}{\partial T} = 3N\hbar\omega\frac{\partial}{\partial T}\left(\frac{1}{e^{\frac{\hbar\omega}{k_BT}}-1}\right)$$

$$= 3N\hbar\omega\frac{\partial}{\partial T}\left(\exp\left(\frac{\hbar\omega}{k_BT}\right)-1\right)^{-1}$$

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$$C_v = 3N\hbar\omega\,(-1)\left(\exp\left(\frac{\hbar\omega}{k_BT}\right)-1\right)^{-2}\frac{\hbar\omega}{k_B}(-1)T^{-2}\exp\frac{\hbar\omega}{k_BT}$$

$$C_v = \frac{3N\hbar\omega\, e^{\frac{\hbar\omega}{k_BT}}\left(\frac{\hbar\omega}{k_B}\right)\frac{1}{T^2}}{\left(e^{\frac{\hbar\omega}{k_BT}}-1\right)^2}$$

$$C_v = \frac{3Nk_B\left(\frac{\hbar\omega}{k_BT}\right)^2 e^{\frac{\hbar\omega}{k_BT}}}{\left(e^{\frac{\hbar\omega}{k_BT}}-1\right)^2}$$

Put $\theta_E = \frac{\hbar\omega}{k_B}$ (called Einstein Temperature)

and $x = \frac{\hbar\omega}{k_BT} = \frac{\theta_E}{T}$

$$C_v = 3R\,\frac{(x)^2 e^x}{(e^x-1)^2}$$

$$C_v = 3R\,E(x)$$

Where $E(x) = \frac{(x)^2 e^x}{(e^x-1)^2}$ (called Einstein function)

The Einstein specific heat curve is fairly close to experimental curve except at low temperature. In fig Einstein specific heat approaches zero more rapidly than obsered value. However at high temperature it approaches

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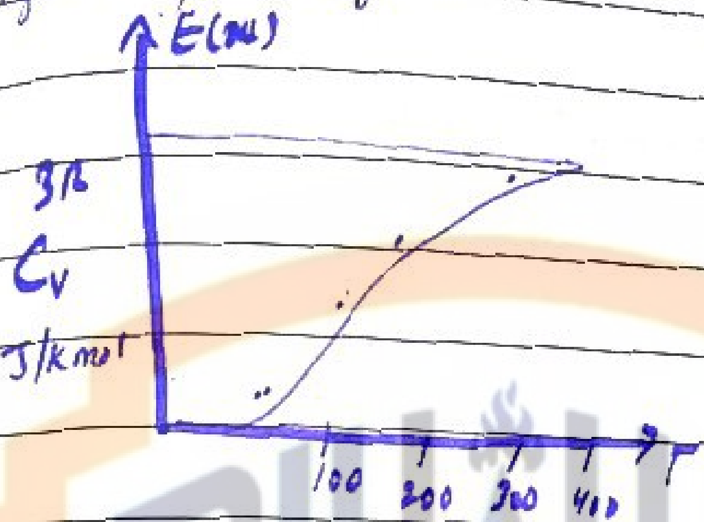
classical value.

Let us consider the two temperature ranges separately.



High Temperature Range:-

$$K_B T >>> 1 \quad , \quad \frac{\hbar\omega}{K_B T} <<< 1$$

$$e^{x} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \text{---}$$

$$e^{x} \simeq 1 + x \quad , \quad \text{let } x = \frac{\hbar\omega}{K_B T}$$

$$e^{\frac{\hbar\omega}{K_B T}} = 1 + \frac{\hbar\omega}{K_B T}$$

$$C_V = 3NK_B \frac{x^2 e^{x}}{(e^{x}-1)^2}$$

For $x <<< 1$, $C_V = 3NK_B \frac{(x^2)(1+x)}{(1+x-1)^2}$

$$C_V = 3NK_B \frac{x^2(1+x)}{x^2}$$

$$C_V = 3NK_B(1+x)$$

$x \to 0$

$$C_V = 3NK_B + 3NK_B x$$

$$C_V \simeq 3NK_B$$

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$C_v = 3R$ which is same as Dulong & Petits laws.

Low temperature Range.

$K_BT <<< 1 \quad \frac{\hbar\omega}{K_BT} >>> 1$

$$\left(e^{\frac{\hbar\omega}{K_BT}} - 1\right) \approx e^{\frac{\hbar\omega}{K_BT}}$$

$$C_v = \frac{3NK_B\left(\frac{\hbar\omega}{K_BT}\right)^2 e^{\frac{\hbar\omega}{K_BT}}}{\left(e^{\frac{\hbar\omega}{K_BT}}\right)^2}$$

$$C_v = 3NK_B\left(\frac{\hbar\omega}{K_BT}\right)^2 e^{-\frac{\hbar\omega}{K_BT}}$$

This equation indicates that the exponential term is more important than term $\left(\frac{\hbar\omega}{K_BT}\right)^2$ in determining the variation of C_v with temperature. By decreasing temperature, C_v drops exponentially the reason for decrease in specific heat at low temperature is that at low temperature most of the oscillators are in the state of zero point energy i.e. $\frac{1}{2}\hbar\omega$ and the

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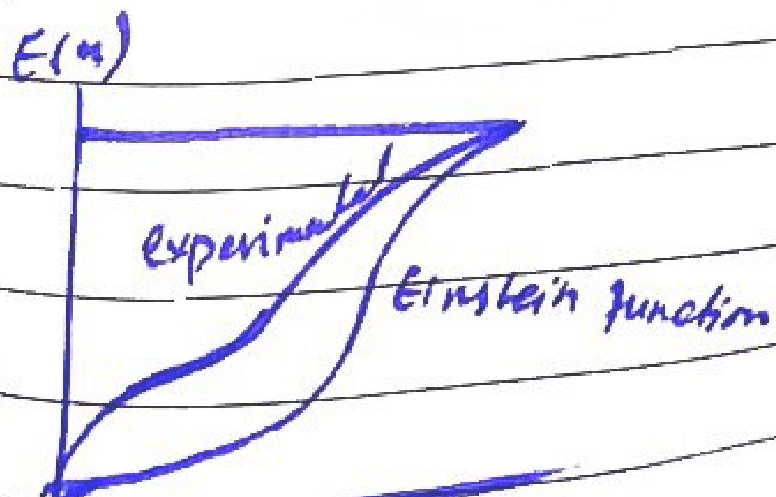
thermal energy require to excite the oscillator is very much less than the quantum energy required to excite the oscillators and therefore the theory fails

Graphic Explanation.

The Einstein function with experimental curve is shown.

It is seen that at high temperature two curve coincides but Einstein function falls off rapidly at low temperature than it should be. Hence Einstein theory is also insufficient to explain T^3 law of Specific heat.



Question 3 (b)

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(b) For a cubic lattice, calculate distance of (123) & (234) planes from a plane passing through origin

$$d = \frac{|c_1 h_1 + c_2 k_1 + c_3 l_1|}{\sqrt{h_1^2 + k_1^2 + l_1^2}} \quad OR$$

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

- where a is lattice constant (distance b/w two nearest lattice points)
- h, k, l are the miller indices of the plane

⇒ For (123) plane $(h,k,l) = (1,2,3)$

$$d_{123} = \frac{a}{\sqrt{(1)^2 + (2)^2 + (3)^2}}$$

$$= \frac{a}{\sqrt{1+4+9}} = \frac{a}{\sqrt{14}}$$

→ For (234) plane

Miller indices $(h, k, l) = (2, 3, 4)$

$$d_{234} = \frac{a}{\sqrt{2^2 + 3^2 + 4^2}}$$

$$= \frac{a}{\sqrt{4 + 9 + 16}} = \frac{a}{\sqrt{29}}$$

→ What is the procedure to find out the

Electrostatic or Madelung Energy

"One atom in the solid-state molecule can have overlapping integrals not only with the nearest-neighbor atoms but also with the second, third, fourth and even with the nth atoms to obtain collective stabilization energy, called Madelung energy."

⇒ The calculation of lattice energy and cohestive energy of ionic crystals was made by **Born and Madelung**.

According to this theory:-

let U_{ij} be interaction energy between ions i and j and U_i the interaction energy on the ion i due to all other ions, then

$$U_i = \sum_{j \neq i} U_{ij}$$

Here, the energy U_i is ~~const~~ consists of two parts:-

(a) A central field repulsive potential of the form $\lambda e^{-r/p}$, where 'r' and 'p' are empirical parameters.

(b) A coulomb potential $\pm q^2/r$.

So,

$$U_{ij} = \lambda e^{-r_{ij}/p} \pm q^2/r_{ij}$$

Here $\because r_{ij} = R$

$$U_{ij} = \lambda e^{-R/p} \pm q^2/R$$

The total lattice energy of the crystal compound of N molecules is:-

$U_{total} = N U_i$

$$U_{total} = N \sum U_{ij}$$

$$U_{total} = N\left(z\lambda e^{-R/p} - \sum \frac{(\pm 1)}{p_{ij}} \frac{q^2}{R}\right)$$

Here, 'z' is the number of nearest neighbor and $\sum \frac{(\pm 1)}{p_{ij}} = \alpha$ is Madelung constant.

$$U_{total} = Nz\lambda e^{-R/p} - \frac{N\alpha q^2}{R} \longrightarrow ⊛$$

At equilibrium seperation $R = r_0$, the crystal potential energy is minimum.

$$\because \left(\frac{dU_{total}}{dR}\right)_{R=r_0} = 0$$

$$= -\frac{Nz\lambda}{p} e^{-R/p} + \frac{N\alpha q^2}{R^2}$$

$$\left(\frac{dU_{total}}{dR}\right)_{R=r_0} = -\frac{Nz\lambda}{p} e^{-r_0/p} + \frac{N\alpha q^2}{r_0^2} = 0 \longrightarrow (1)$$

2

$$-\frac{Nz\lambda}{p} e^{-r_0/p} = -\frac{N\alpha q^2}{r_0^2}$$

$$e^{-r_0/p} = \frac{p\alpha q^2}{r_0^2 z\lambda}$$

Putting above value in ★.

$$U_{total} = Nz\lambda e^{-r_0/p} - \frac{N\alpha q^2}{R}$$

$$= N\cancel{z}\cancel{\lambda} \times \frac{p\alpha q^2}{r_0^2 \cancel{z}\cancel{\lambda}} - \frac{N\alpha q^2}{R}$$

$$= \frac{Np\alpha q^2}{r_0^2} - \frac{N\alpha q^2}{R}$$

$$U_{total} = -\frac{N\alpha q^2}{r_0}\left(-\frac{p}{r_0} + 1\right)$$

$$U_{total} = -\frac{N\alpha q^2}{r_0}\left(1 - \frac{p}{r_0}\right)$$

Here, the term $-\frac{N\alpha q^2}{r_0}$ is called Madelung energy.

Thus, $U_{total} = U_{madelung}\left(1 - \frac{p}{r_0}\right)$

Evaluation of Madelung Constant.

$$U_{total} = -\frac{N\alpha q^2}{r_0}\left(1 - \frac{p}{r_0}\right)$$

Here 'α' is known as Madelung Constant.

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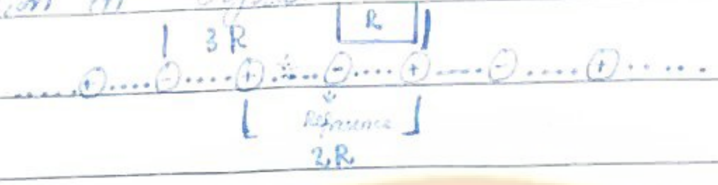
$$\alpha = \sum_j{}' \frac{(\pm 1)}{p_{ij}}$$

$\because$ p_{ij} is a dimensionless quantity as is given by the relation $r_{ij} = p_{ij} R$ $\because$

R is being nearest neighbor seperation in crystal.



If we take refrence ion as negative charge, then plus sign will be used for positive ion and negative sign will be used for negative ion.

$$\frac{\alpha}{R} = \sum_j{}' \frac{(\pm 1)}{R\, p_{ij}} = \sum{}' \frac{(\pm 1)}{r_j}$$

For one dimension:- (NaCl)

$$\frac{\alpha}{R} = 2\left\{1 - \frac{1}{2R} + \frac{1}{3R} - \frac{1}{4R} + \dots\dots\right\}$$

$$\alpha = 2\left\{1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots\dots\right\} \longrightarrow (M)$$

The factor 2 has been taken because there are two ions, one to the right and one to the left of refrence ion.

As we know the bracket series

$$\because \ln(1+x) = x - \frac{x}{2} + \frac{x}{3} - \frac{x}{4} + \dots\dots$$

if

$$\ln(1+1) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots\dots$$

4

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots\ldots \rightarrow (N)$$

By comparing eq (M) and (N).

$$\boxed{\alpha = 2\ln 2}$$

So, Madelung constant is $2\ln 2$.

$$\boxed{\alpha = 2 \times 6.9 \Rightarrow 1.38}$$

For Two-Dimensions. (NaCl)

$$\alpha = 4 - \frac{4}{\sqrt{2}} + \frac{4}{2} - \frac{8}{2\sqrt{2}} + \ldots\ldots\ldots$$

For Three-Dimensions:- (NaCl)

$$\alpha = \frac{6}{\sqrt{1}} - \frac{12}{\sqrt{2}} + \frac{8}{\sqrt{3}} - \frac{6}{\sqrt{4}} + \ldots\ldots$$

$$\boxed{\alpha = 1.748}$$

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Example:-

	Ne	Ar	Kr	Xe
$\frac{R_0}{\sigma}$	1.14	1.11	1.10	1.09

long Question 1(b)

Cohesive Energy

"The energy required to seperate atoms apart from each other and to bring them to an assembly of neutral free atoms, at rest or at infinite seperation."

- Noble gases have small cohesive energies. may be zero.

Derievation:-

The total potential energy for atom is

$$U_{tot}(R) = 2N\epsilon\left[12.13\left(\frac{\sigma}{R}\right)^{12} - 14.45\left(\frac{\sigma}{R}\right)^{6}\right]$$

Put $R = R_\circ$

$$U_{total}(R_\circ) = 2N\epsilon\left[12.13\left(\frac{\sigma}{R_\circ}\right)^{12} - 14.45\left(\frac{\sigma}{R_\circ}\right)^{6}\right]$$

AS:-

$$\boxed{R_\circ/\sigma = 1.09}$$

So,

$$U_{total}(R_\circ) = 2N\epsilon\left[12.13\left(\frac{1}{1.09}\right)^{12} - 14.45\left(\frac{1}{1.09}\right)^{6}\right]$$

$$= -8.15\,N\epsilon \quad \text{or}$$

$$U_{total}(R) = (-2.15)(4N\epsilon)$$

The cohesive energy of inert gas crystal at

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absolute zero and zero presurre. It is same for all inert gases, when K.E is zero i.e the atoms are at rest.